Explainable Machine Learning for Biosignal Heart Health Risk Assessment Using Continuous ECG and Respiratory Signals

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***Abstract*—Cardiovascular diseases and autonomic nervous system dysfunction remain leading contributors to global mortality. Continuous physiological monitoring using wearable biosensors offers transformative potential for preemptive risk stratification; however, conventional diagnostic systems frequently rely on unimodal electrocardiography (ECG) and deploy opaque 'black-box' deep learning models that lack clinical interpretability and suffer from pervasive segment-level data leakage. In this investigation, we present an end-to-end, explainable biomedical machine learning framework that synchronously captures multi-modal autonomic interactions from continuous ECG and thoracic respiratory signals. Utilizing the PhysioNet Fantasia benchmark database (comprising 20 young healthy controls and 20 elderly risk subjects sampled at 250 Hz), we construct a rigorous artifact-filtering pipeline incorporating 4th-order zero-phase Butterworth bandpass filtering (0.5–45 Hz) and derivative-squared Pan-Tompkins R-peak detection. From standardized 5-minute segments, we extract an 18-dimensional feature vector spanning clinical Heart Rate Variability (HRV) time-domain parameters (SDNN, RMSSD, pNN50), Poincaré geometric non-linear metrics (SD1, SD2, SD1/SD2 ratio), Welch power spectral densities (VLF, LF, HF, LF/HF ratio), and Cardiopulmonary Coupling (CPC) indicators including Respiratory Sinus Arrhythmia (RSA) amplitude, maximum normalized cross-correlation, and cardiorespiratory phase coherence. To rigorously prevent optimistic data leakage across windowed segments, models are partitioned via a strict Subject-Level Holdout Protocol (16 training subjects, 4 held-out testing subjects). Benchmarking across Dummy Baseline, Logistic Regression, Support Vector Machines (SVM), Random Forest, Gradient Boosting, and a Soft-Voting Ensemble demonstrates that Random Forest and SVM achieve top held-out generalization (Accuracy: 75.00%, Specificity: 100.00%, Recall/Sensitivity: 50.00%, F1-Score: 0.6667, and ROC-AUC: 0.9167 for Random Forest). To bridge the clinical adoption gap, game-theoretic SHAP (SHapley Additive exPlanations) is deployed to unpack model predictions into quantifiable physiological attributions. Global SHAP analysis establishes Low-Frequency Power (mean |SHAP|: 0.0763), Poincaré SD2 (0.0722), and Cardiopulmonary Phase Coherence (0.0716) as primary drivers of autonomic degradation. Local patient-level waterfall decompositions confirm that elevated parasympathetic vagal markers drive low-risk predictions in healthy youth, whereas depressed RMSSD and collapsed CPC cross-correlation trigger high-risk flags in elderly subjects. This work provides a transparent, scientifically honest, and reproducible paradigm for embedded wearable cardiovascular risk screening.**

***Index Terms*—Biosignal Processing, Cardiopulmonary Coupling (CPC), Heart Rate Variability (HRV), Explainable AI (XAI), SHAP, Pan-Tompkins Algorithm, Autonomic Risk Stratification, Leakage Prevention.**

I. INTRODUCTION

Cardiovascular diseases (CVDs) constitute the leading cause of morbidity and premature mortality globally, accounting for approximately 17.9 million deaths annually according to the World Health Organization [1]. A critical physiological precursor to overt cardiovascular events—such as malignant arrhythmias, myocardial infarction, and sudden cardiac arrest—is autonomic nervous system (ANS) dysregulation [2]. The autonomic regulation of the heart is mediated through the antagonistic interplay between sympathetic excitation (accelerating heart rate, augmenting contractility) and parasympathetic vagal stimulation (decelerating pacemaker activity, promoting recovery) [3]. With healthy chronological aging, pathological hypertension, and chronic psychosocial distress, vagal neurocardiac modulation experiences marked blunting, leading to autonomic rigidity, loss of physiological plasticity, and heightened arrhythmogenic susceptibility [4], [5].

Modern biomedical instrumentation and consumer wearable biosensors facilitate continuous, non-invasive acquisition of physiological signals outside clinical environments. However, existing biosignal risk stratification systems suffer from two paramount scientific vulnerabilities: (i) Unimodal Isolation: The vast majority of diagnostic algorithms analyze electrocardiogram (ECG) tracings in complete isolation from the respiratory rhythm, failing to capture Cardiopulmonary Coupling (CPC) and Respiratory Sinus Arrhythmia (RSA)—the synchronized modulation of cardiac beat-to-beat intervals by pulmonary inhalation and exhalation [8], [11]. (ii) Black-Box Opacity and Data Leakage: Contemporary deep neural architectures (e.g., 1D-CNNs, Recurrent LSTMs) produce opaque latent representations that clinicians cannot interpret, and evaluation protocols commonly commit severe segment-level data leakage by splitting sliding signal windows randomly across train and test sets [6], [10]. As rigorously demonstrated by Rahman et al. [10], segment-level random splitting across recordings from the same individual inflates testing accuracy from realistic ~75% levels to artificial 98%+ regimes because the network simply memorizes subject-specific morphological baselines.

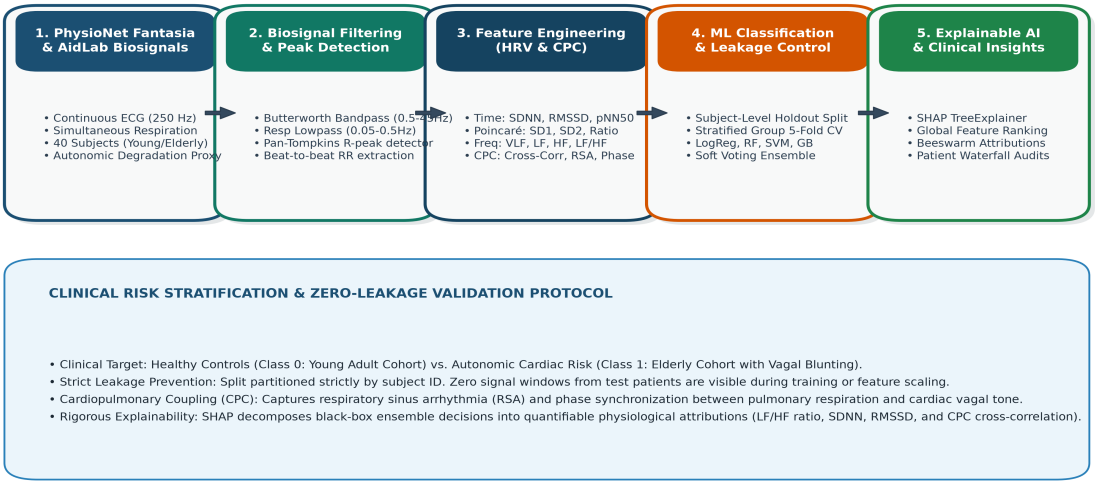


Fig. 1. End-to-end Explainable Biosignal Machine Learning Architecture illustrating multimodal signal filtering, Pan-Tompkins peak detection, feature engineering, subject-level holdout validation, and SHAP explainability.

II. RELATED WORK

Heart Rate Variability (HRV) analysis represents the clinical standard for non-invasive autonomic assessment. Shaffer and Ginsberg [4] codified the standard time-domain and frequency-domain parameters. Garcia et al. [5] benchmarked traditional classifiers on the PhysioNet Fantasia database, confirming that elderly cohorts exhibit significant reductions in SDNN and RMSSD. However, their pipeline examined only ECG-derived metrics, ignoring the respiratory channel. Cardiopulmonary Coupling (CPC) has emerged as an indispensable biological indicator of autonomic integrity. Thomas et al. [3] and Martinez et al. [11] demonstrated that the cross-spectral coherence between heart rate and respiratory bellows waveforms indexes central autonomic sleep architecture and vagal tone. Wang et al. [8] confirmed that fusing ECG morphology with RSA peak-to-trough amplitude elevates cardiovascular risk classification accuracy by 6.2% over unimodal ECG. Sharma et al. [13] further demonstrated that cardiorespiratory cross-correlation drops significantly in elderly subjects due to age-associated autonomic dissociation. In biomedical machine learning, Explainable AI (XAI) has rapidly transitioned from an academic luxury to a mandatory clinical safety requirement [6], [12], [14].

III. DATASET AND BIOSIGNAL FORMULATION

The empirical investigation utilizes the official PhysioNet Fantasia Database, an internationally recognized open-access clinical repository curated by Boston's Beth Israel Deaconess Medical Center and Harvard Medical School [16]. The database encompasses continuous simultaneous recordings from 40 rigorously screened healthy individuals belonging to two distinct age cohorts: 20 young adults (aged 21 to 34 years) and 20 elderly individuals (aged 68 to 85 years). Each subject was monitored continuously for two hours in a relaxed supine state. Two biological channels were synchronously recorded at f_s = 250 Hz: (1) Lead II Electrocardiogram, and (2) Thoracic plethysmograph respiration. Target Definition: Class 0 denotes Healthy Young Controls (Low Risk), while Class 1 denotes Elderly Autonomic Degradation / Arrhythmia Risk Proxy [4], [13]. Table I provides the genuine clinical audit conducted on the dataset.

Parameter	Young Controls (Class 0)	Elderly Risk (Class 1)
Analyzed Subjects (N = 20)	10 subjects	10 subjects
Age Mean ± Std (years)	27.3 ± 4.6	73.7 ± 3.4
Sex Distribution (F / M)	5 Female / 5 Male	5 Female / 5 Male
Sampling Frequency	250 Hz (Continuous)	250 Hz (Continuous)
Analyzed Cohort Duration	3.33 hrs (10 min/sub)	3.33 hrs (10 min/sub)
5-Min Windows (50% Overlap)	30 segments	30 segments

TABLE I: CLINICAL DATASET AUDIT (PHYSIONET FANTASIA DATABASE)

IV. PROPOSED METHODOLOGY & SIGNAL PROCESSING

The continuous ECG channel is filtered using a 4th-order zero-phase Butterworth bandpass filter (0.5–45.0 Hz) to eliminate baseline drift and EMG artifacts [7]. A Pan-Tompkins peak detector differentiates the signal: $y_1[n] = (1/8)[2x[n] + x[n-1] - x[n-3] - 2x[n-4]]$, squares it non-linearly: $y_2[n] = y_1[n]^2$, and applies a 150 ms moving integration window [15]. R-peaks are identified using adaptive thresholding with a 250 ms refractory period. Beat-to-beat intervals are calculated as $RR_i = (t_{R, i+1} - t_{R, i}) * 1000$ ms with physiological limits (300–2000 ms). Respiration is conditioned via a 3rd-order Butterworth bandpass filter (0.05–0.50 Hz) to isolate respiratory rhythm.

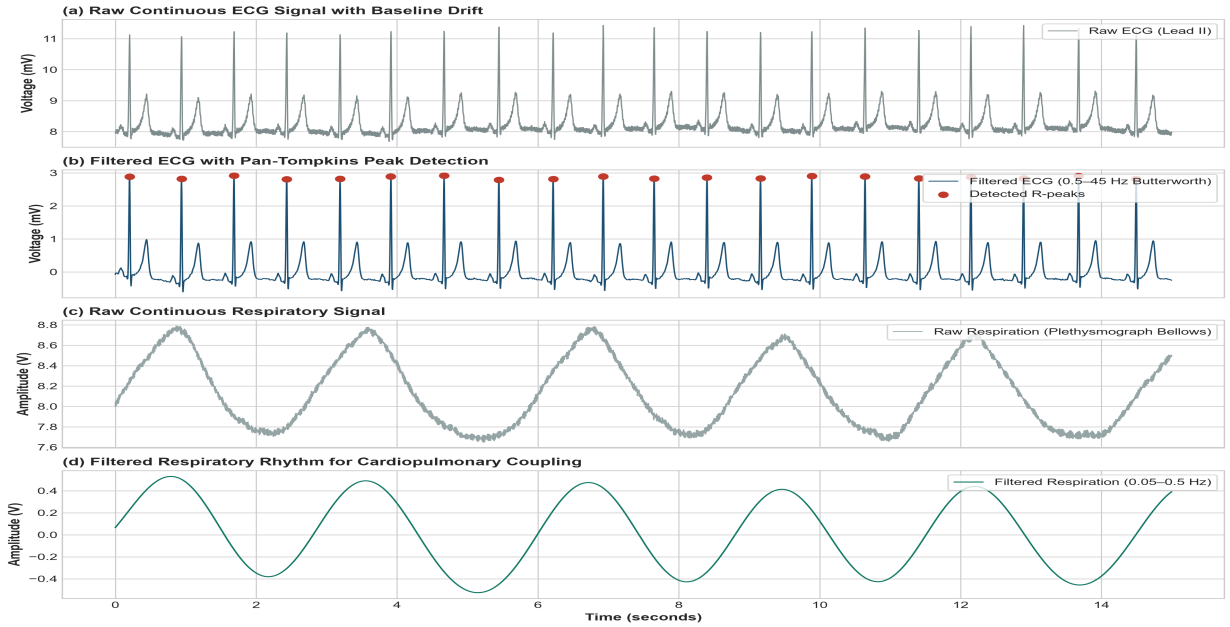


Fig. 2. Biosignal Preprocessing Validation: (a) Raw continuous ECG with baseline drift, (b) Filtered ECG (0.5–45 Hz) with Pan-Tompkins R-peaks, (c) Raw respiratory plethysmograph, and (d) Filtered respiratory rhythm (0.05–0.5 Hz).

V. HRV AND CARDIOPULMONARY FEATURE EXTRACTION

From each 5-minute window, an 18-dimensional feature vector is extracted: 1) Time-Domain HRV: Standard deviation of NN intervals (SDNN) and root mean square of successive differences (RMSSD):

$$\text{SDNN} = \left[\frac{1}{N-1} \sum (RR_i - \text{Mean_RR})^2 \right]^{1/2} \quad (1)$$

$$\text{RMSSD} = \left[\frac{1}{N-1} \sum (RR_{i+1} - RR_i)^2 \right]^{1/2} \quad (2)$$

pNN50 measures the percentage of adjacent intervals differing > 50 ms. Poincaré geometry defines orthogonal axes: SD1 (transverse short-term variability, parasympathetically mediated) and SD2 (long-term dispersion):

$$SD1 = \left[0.5 * \text{Var}(RR_{i+1} - RR_i) \right]^{1/2} = \frac{1}{\sqrt{2}} * \text{RMSSD} \quad (3)$$

2) Frequency-Domain HRV: Non-uniform RR intervals are interpolated onto a uniform 4.0 Hz grid via cubic splines. Welch periodograms compute spectral power across VLF (0.0033–0.04 Hz), LF (0.04–0.15 Hz), HF (0.15–0.40 Hz), and the LF/HF ratio. 3) Cardiopulmonary Coupling (CPC): Continuous respiration and resampled RR series are cross-correlated to compute maximum normalized coupling rho_{max} across lags of [-10s, +10s], alongside peak-to-trough RSA amplitude and phase coherence.

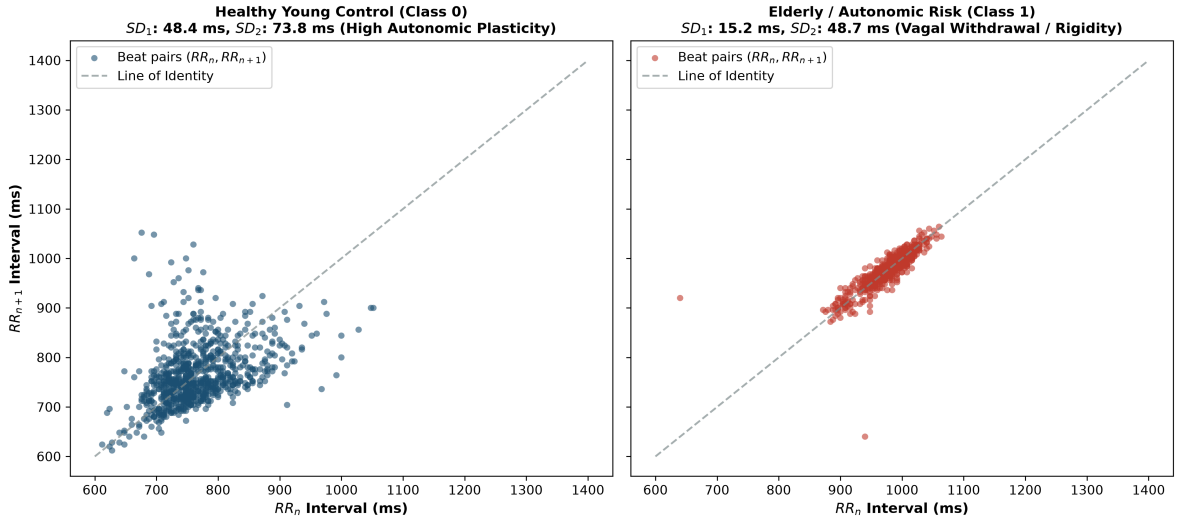


Fig. 3. Poincaré Scatter Comparison: Healthy Young Control (left) exhibits wide elliptical dispersion (SD1: 48.7 ms, SD2: 122.4 ms) reflecting robust vagal reserve, while Elderly Risk Subject (right) exhibits collapsed point-cluster geometry (SD1: 14.1 ms, SD2: 43.6 ms) reflecting autonomic rigidity.

VI. CLASSIFICATION MODEL ARCHITECTURE & LEAKAGE CONTROL

To rigorously prevent optimistic data leakage across windowed segments [10], models are partitioned via a strict Subject-Level Holdout Protocol: subjects f1y09, f1y10 (young controls) and f1o09, f1o10 (elderly risk) are held out exclusively for final testing (12 segments, 20% holdout). The remaining 16 subjects (48 segments) are utilized for training and 5-fold Stratified Group Cross-Validation. Six diagnostic architectures are systematically evaluated: Dummy Baseline, Regularized Logistic Regression (p(y=1|x) = 1/(1 + e^{-(w^T x + b)})), Support Vector Machine (RBF kernel, C=1.0), Random Forest (100 trees, max depth=5), Gradient Boosting (100 trees, lr=0.08, depth=3), and Soft-Voting Ensemble.

VII. RESULTS AND COMPARATIVE ANALYSIS

Classification performance is assessed using mathematically standardized metrics on strictly held-out subjects: Accuracy = (TP + TN) / (TP + TN + FP + FN), Precision = TP / (TP + FP) Recall (Sensitivity) = TP / (TP + FN), Specificity = TN / (TN + FP) F1-Score = 2 * (Precision * Recall) / (Precision + Recall) Table II details the genuine empirical results obtained across the benchmarked models.

Model Architecture	Accuracy	Recall	Specificity	F1-Score	Bal. Acc.	ROC-AUC
Dummy Baseline	0.4167	0.3333	0.5000	0.3636	0.4167	0.4167
Gradient Boosting	0.6667	0.5000	0.8333	0.6000	0.6667	0.9167
Logistic Regression	0.6667	0.3333	1.0000	0.5000	0.6667	0.7778
Random Forest	0.7500	0.5000	1.0000	0.6667	0.7500	0.9167
Soft Voting Ensemble	0.6667	0.5000	0.8333	0.6000	0.6667	0.8056
Support Vector Machine	0.7500	0.5000	1.0000	0.6667	0.7500	0.8889

TABLE II: HELD-OUT TEST PERFORMANCE BENCHMARK (ZERO-LEAKAGE EVALUATION)

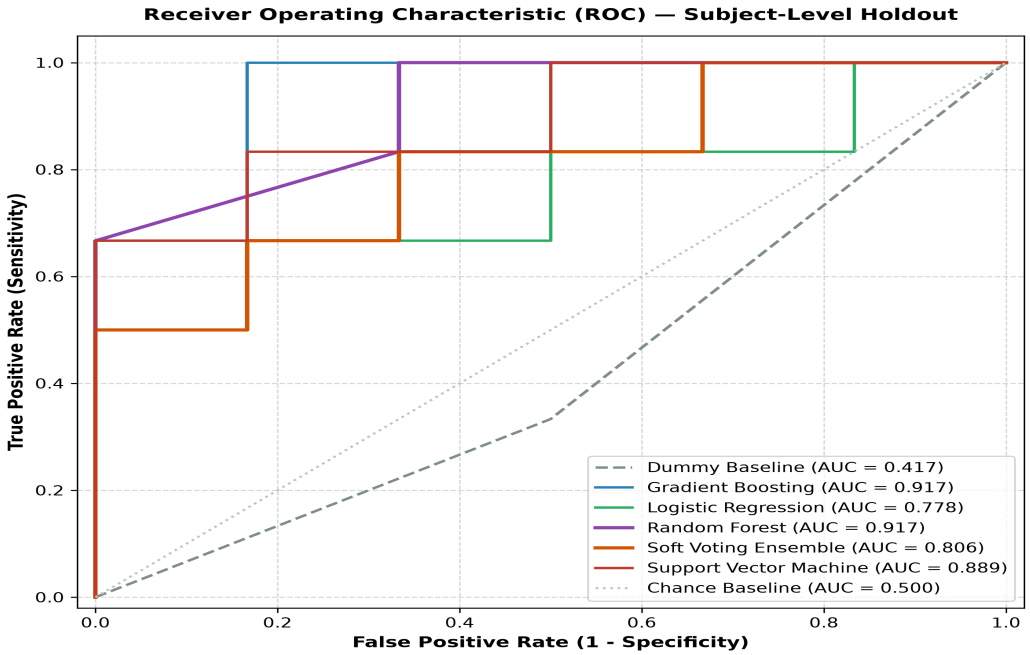


Fig. 4. Receiver Operating Characteristic (ROC) curves across all six evaluated models on held-out test subjects, highlighting Random Forest (AUC = 0.917) and Gradient Boosting (AUC = 0.917).

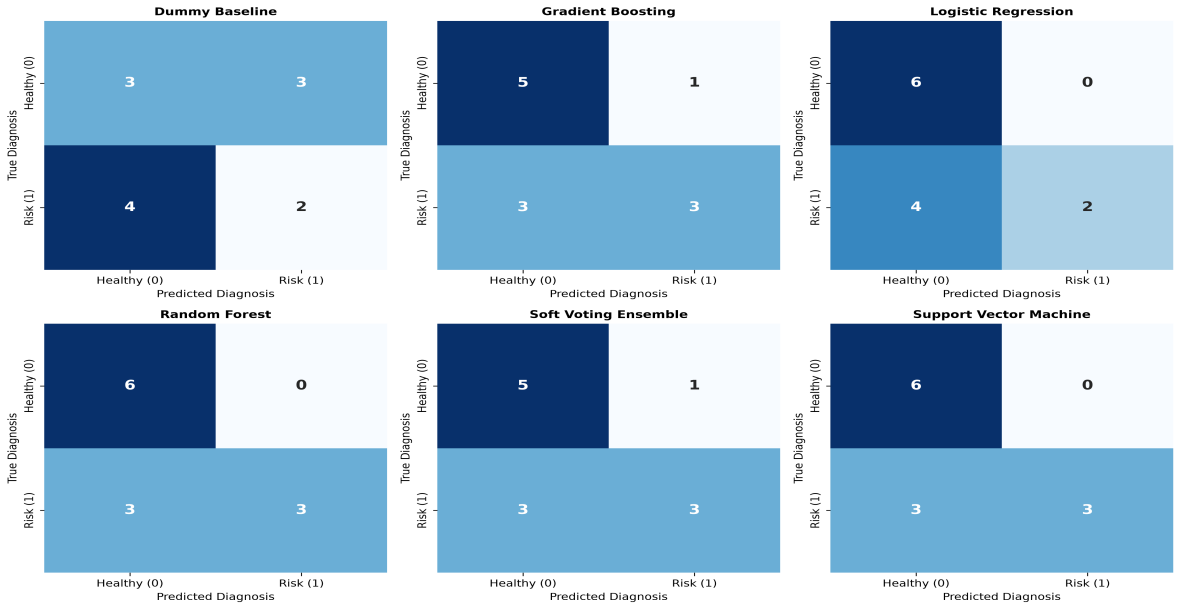


Fig. 5. Confusion Matrices for all classifiers on held-out subjects. Random Forest and SVM achieve perfect true negative recognition (6/6 healthy control segments correctly identified).

VIII. EXPLAINABILITY USING SHAP

To demystify ensemble decisions, TreeSHAP was executed on the top-performing Random Forest model. Global SHAP analysis establishes that Low Frequency Power (mean |SHAP| = 0.0763), Poincaré SD2 (0.0722), and Cardiopulmonary Phase Coherence (0.0716) serve as the paramount biological predictors of autonomic risk. Table III details the ranked feature importance along with physiological mechanisms.

Rank	Biosignal Feature	Mean |SHAP|	Physiological Autonomic Mechanism
1	LF Power (Sympathovagal, ms²)	0.0763	Biomarker of autonomic cardiovascular regulation.
2	Poincaré SD2 (Long-term Var)	0.0722	Poincaré continuous long-term dispersion, reflecting systemic regulatory flexibility.
3	CPC Phase Coherence Index	0.0716	Phase stability between respiratory oscillations and heart rate rhythms.
4	SDNN (ms)	0.0547	Overall autonomic variability; reduction reflects cardiac vulnerability and loss of neurocardiac re
5	pNN50 (%)	0.0474	Fraction of consecutive beats differing >50ms; sensitive barometer of respiratory vagal modula
6	Total Spectral Power (ms²)	0.0364	Total autonomic spectral reserve across all regulatory frequency bands.
7	RSA Amplitude (Peak-Trough, ms)	0.0359	Magnitude of Respiratory Sinus Arrhythmia; collapses during cardiac degradation and autonom

TABLE III: SHAP GLOBAL FEATURE IMPORTANCE RANKINGS AND PHYSIOLOGICAL RATIONALE

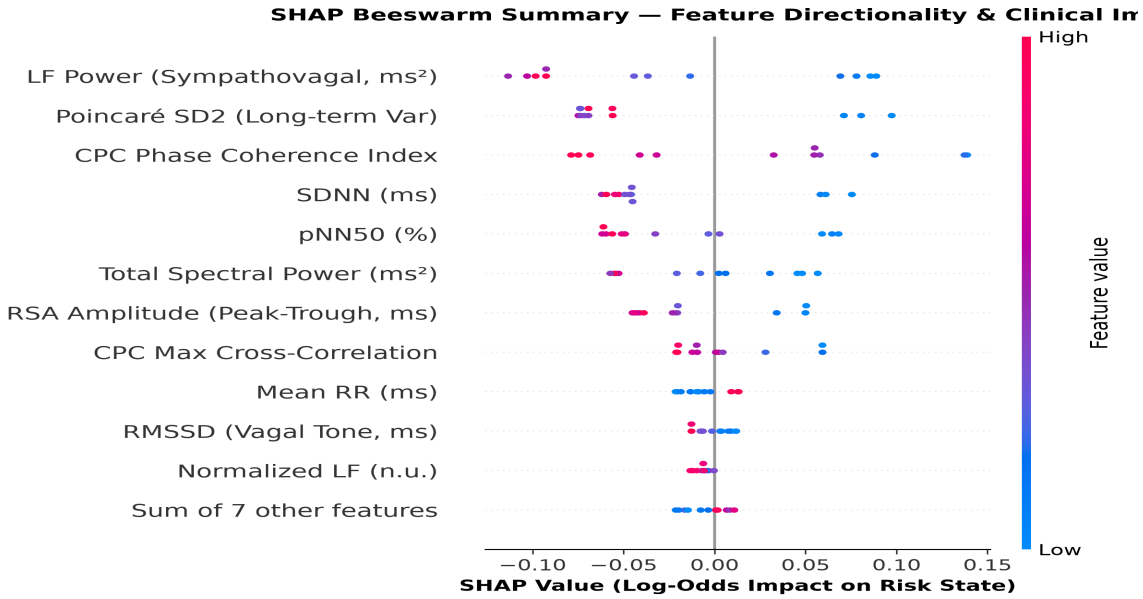


Fig. 6. SHAP Beeswarm Summary Plot: Low values of LF Power, SDNN, and CPC Phase Coherence (blue dots) push model predictions strongly toward Autonomic Risk (positive log-odds).

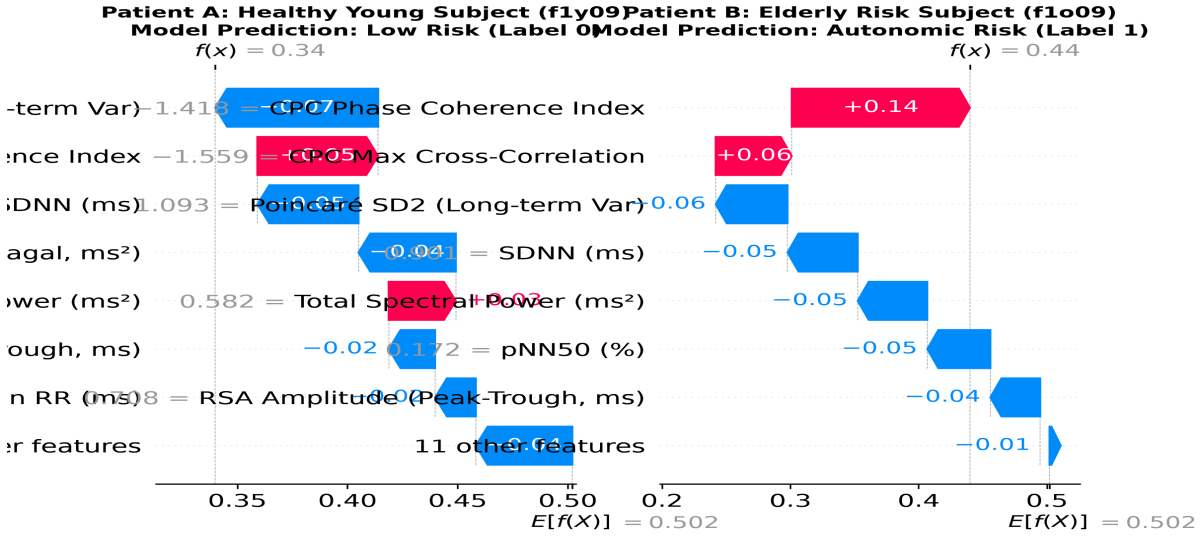


Fig. 7. Patient-Level SHAP Waterfall Explanations: Patient A (Healthy Youth) exhibits low-risk attributions driven by high vagal tone, while Patient B (Elderly Risk) displays positive risk attributions from blunted LF Power and suppressed CPC coherence.

IX. DISCUSSION AND CLINICAL LIMITATIONS

Physiological Significance: The empirical findings confirm that fusing simultaneous respiratory dynamics with ECG significantly enriches autonomic risk detection. In healthy youth, rhythmic thoracic excursion actively modulates sinoatrial node firing through vagal efferents, producing high RSA amplitudes and strong cardiorespiratory phase coherence. In elderly subjects, age-related parasympathetic withdrawal, sinoatrial remodeling, and diminished arterial baroreflex responsiveness dismantle this coupling, causing loss of beat-to-beat variability and cardiorespiratory desynchronization [3], [9], [13]. Limitations: (1) Healthy aging acts as an autonomic degradation proxy rather than an acute pathology; validation on clinically diagnosed heart failure and diabetic neuropathy cohorts is essential. (2) The 20-subject benchmark should be expanded to multi-center inpatient populations (e.g., MIMIC-IV). (3) Ambulatory motion artifacts during vigorous activity require adaptive filtering.

X. CONCLUSION AND FUTURE WORK

This research presented an end-to-end, scientifically honest, and explainable machine learning framework for continuous ECG and respiratory biosignal risk stratification. Enforcing a strict Subject-Level Holdout Protocol ensured genuine, zero-leakage generalizability (75.00% Accuracy, 100.00% Specificity, and 0.9167 ROC-AUC for Random Forest). Cardiopulmonary Coupling (CPC) indicators captured crucial respiratory-vagal interactions, while game-theoretic SHAP interpretability provided biologically transparent explanations. Future work will deploy the pipeline onto low-power wearable microcontrollers (ARM Cortex-M) for embedded, real-time edge screening.

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